

Ramanujan Madhusudhan

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[Linkedin](#)

[Project Portfolio](#)

Education

Bachelor of Engineering (Hons) - Computer Systems Engineering - 3rd Year - University of Auckland

Experience

21e8 - (AI Labs Startup)

Software Intern (July 2025, Nov 2025 - Feb 2026)

- Developed full-stack software for managing and exploring geospatial drone data for a commercial drone client.
- Built client-side applications using Svelte, HTML, CSS and JavaScript based on user stories and different user personas
- Developed backend functionality in C++ using http lib and Taskflow implementing REST-like APIs.
- Worked with SQLite, containerisation, Git, GitHub, Auth0 and MinIO/S3
- Communicated project progress to the client through regular video presentations.

Electronic Partners

Production Assembly, Testing and House-Keeping Engineer (Nov 2025 - Feb 2026)

- Tested and debugged PCBs for various clients including lift control systems.
- Managed electronic component inventory and prepared BOMs for PCB assembly.
- Developed a file transfer system between Android devices and Microcontrollers.

Projects

Auckland Programme for Space Systems (APSS)

Avionics Team Lead (July 2024 - Present)

- Lead a team of undergraduate engineers responsible for CubeSat avionics development, testing and system integration.
- Designed and tested a radiation detector PCB for the APSS-3 CubeSat using analogue signal processing and a silicon photomultiplier.
- Contributed to the mechanical and thermal-vacuum qualification testing of APSS-2, including sensor calibration and debugging payload-board test software.
- Developed a battery test plan for CubeSat flight-readiness testing.
- Collaborate with multidisciplinary engineering teams to integrate avionics with other CubeSat subsystems.

Car Charging Console

- Developed a cloud-connected car charging management application using Svelte, C#, Azure Functions and Azure IoT Hub.
- Implemented REST API endpoints to start and stop charging and configure charging schedules.
- Connected the web application to a simulated IoT device through Azure IoT Hub.

Additional Projects: Built a smart home system using Home Assistant; deployed and remotely managed a containerised NAS on Linux using SSH.

Indie Game Developer (2019-2022)

Developed games through game jams using Unity, C#, level design, animation and visual effects.

[Itch.io](https://itch.io)

Skills

- **Programming Languages:** C, C++, C#, Python, Java, JavaScript
- **Web & Software:** Svelte, HTML, CSS, REST APIs, SQLite
- **Cloud & Systems:** Linux, Bash, SSH, Docker/Containerisation, Azure Functions, Azure IoT Hub
- **Embedded & Hardware:** Altium Designer, PCB Design, VHDL, FPGA, Hardware Testing
- **Tools:** Git, GitHub, Unity